

Problem-Solution Fit

Date	15 March 2026
Team ID	SWTID-2026-2085
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Small-scale farmers, large-scale farmers, agricultural consultants, and farming communities seeking better crop recommendations.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Limited budget, lack of technical knowledge, limited internet access, and basic smartphones or computers.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Traditional farming advice, agricultural experts, and online resources. Pros: Experienced guidance. Cons: Time-consuming, costly, and not personalized.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Farmers often struggle to select the right crop due to changing weather, poor soil knowledge, and lack of expert guidance. These	9. PROBLEM ROOT / CAUSE [RC] Lack of accurate crop recommendations based on soil nutrients and environmental conditions, leading to poor crop selection and lower yields.	7. BEHAVIOR + ITS INTENSITY [BE] Farmers frequently seek advice from experts, neighbors, or online sources before every planting season and are willing to adopt reliable technology.

issues occur every farming season

3. TRIGGERS TO ACT

[TR] Need to improve crop yield, reduce financial losses, and make better farming decisions using data-driven insights.

4. EMOTIONS

BEFORE / AFTER [EM]

Before: Confused, worried, uncertain. **After:** Confident, satisfied, hopeful about higher productivity.

10. YOUR SOLUTION

[SL] OptiCrop uses machine learning to analyze soil nutrients and environmental data and recommends the most suitable crop, helping farmers improve productivity, reduce losses, and make informed decisions.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE: Web applications, agriculture websites, YouTube, weather services.

OFFLINE : Agricultural officers, local markets, farmer groups, training programs.